Sets, Relations and Functions

From the class book: Chapter 1, Sections 1.1 to 1.4

1. Key definitions

Power set.

- $P(A)$ is the set of all subsets of A.
- If $|A| = n$ then $|P(A)| = 2^n$.

Symmetric difference.

- $A \triangle B = (A - B) \cup (B - A)$.
- The pair $(P(X), \triangle)$ is an abelian group with identity $\emptyset$, and every element is its own inverse.

Relation. A relation from A to B is any subset of $A \times B$; a relation on A is a subset of $A \times A$.

A relation R on A is called

- **reflexive** if $(a, a) \in R$ for every $a \in A$; **irreflexive** if $(a, a) \notin R$ for every a;
- **symmetric** if $(a, b) \in R \Rightarrow (b, a) \in R$;
- **antisymmetric** if $(a, b), (b, a) \in R \Rightarrow a = b$; **asymmetric** if $(a, b) \in R \Rightarrow (b, a) \notin R$;
- **transitive** if $(a, b), (b, c) \in R \Rightarrow (a, c) \in R$.

Equivalence relation.

- A relation that is reflexive, symmetric and transitive.
- The equivalence class of a is $[a] = \{x : x \sim a\}$.

Partial order. A relation that is reflexive, antisymmetric and transitive (for example $\leq$ on $\mathbb{R}$, divisibility on $\mathbb{N}$).

Partition. A family of non-empty, pairwise disjoint subsets of A whose union is A.

Function.

- $f : A \to B$ assigns to each $a \in A$ exactly one $f(a) \in B$.
- It is *one-one* (injective) if $f(x) = f(y) \Rightarrow x = y$, *onto* (surjective) if $f(A) = B$, and a *bijection* if it is both.

2. Main results

Set identities that are often tested

- $P(A) \cap P(B) = P(A \cap B)$, but in general only $P(A) \cup P(B) \subseteq P(A \cup B)$.
- $A - B = B - A \iff A = B$; $A \cup B = A \cap B \iff A = B$; $A - (A - B) = A \cap B$.
- $(X \times Y) \cap (Y \times X) = (X \cap Y) \times (X \cap Y)$, which has $|X \cap Y|^2$ elements.
- Inclusion and exclusion: $|A \cup B \cup C| = \sum |A| - \sum |A \cap B| + |A \cap B \cap C|$.

Theorem (Equivalence relations and partitions)

- The equivalence classes of an equivalence relation on A form a partition of A, and conversely every partition of A comes from exactly one equivalence relation ("same block").
- Two classes are either equal or disjoint, and $[a] = [b] \iff a \sim b$.

Consequently, the number of equivalence relations on an n-set equals the number of partitions, the **Bell number** B_n; those with exactly k classes number $S(n,k)$, the **Stirling number of the second kind**.

Bell and Stirling numbers

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k, \qquad B_n = \sum_{k=1}^{n} S(n,k), \qquad S(n,2) = 2^{n-1} - 1, \qquad S(n,n-1) = \binom{n}{2}.$$

n	0	1	2	3	4	5	6	7
B_n	1	1	2	5	15	52	203	877

- Bell triangle: begin each row with the last entry of the previous row; each further entry is its left neighbour plus the entry directly above that neighbour.
- Rows: 1; 1, 2; 2, 3, 5; 5, 7, 10, 15; …
- The first column reads 1, 1, 2, 5, 15, 52, …

Proposition (Composition)

Let $f : A \to B$ and $g : B \to C$.

- $g \circ f$ one-one $\Rightarrow f$ one-one; $g \circ f$ onto $\Rightarrow g$ onto.
- A composition of bijections is a bijection, and f is a bijection iff it has an inverse.
- If A is finite, a map $A \to A$ is one-one $\iff$ it is onto.

- Proof idea: if $f(x) = f(y)$ then $g(f(x)) = g(f(y))$, so $x = y$; and every $c \in C$ equals $g(f(a))$, which lies in $g(B)$.
- The bijections of a set X onto itself form the symmetric group S_X.

3. Counting tables

Relations on an n-element set

- Method: think of the relation as an $n \times n$ 0-1 matrix with n diagonal cells and $\binom{n}{2}$ pairs of off-diagonal cells $\{(a,b),(b,a)\}$.
- Count the choices for each diagonal cell and each off-diagonal pair, then multiply.

Type of relation	Choices (diagonal; each pair)	Number	$n = 3$
all relations	2 per cell	2^{n^2}	512
reflexive	1; 4	2^{n^2-n}	64
irreflexive	1; 4	2^{n^2-n}	64
symmetric	2; 2	$2^{n(n+1)/2}$	64
reflexive and symmetric	1; 2	$2^{n(n-1)/2}$	8
antisymmetric	2; 3	$2^n\, 3^{n(n-1)/2}$	216
asymmetric	1 (empty); 3	$3^{n(n-1)/2}$	27
symmetric and antisymmetric	2; 1 (empty)	2^n	8
equivalence relations	no product rule	B_n	5
partial orders	no product rule	1, 3, 19, 219 for $n = 1, \ldots, 4$	19

From A to B with $|A| = m$, $|B| = n$ there are 2^{mn} relations.

Functions $A \to B$, $|A| = m$, $|B| = n$

Type	Number	$m = 4,\ n = 3$
all functions	n^m	81
one-one (needs $m \le n$)	$n(n-1)\cdots(n-m+1)$	0
onto (needs $m \ge n$)	$n!\, S(m,n) = \sum_{k=0}^{n}(-1)^k \binom{n}{k}(n-k)^m$	36
bijections ($m = n$)	$n!$	none
binary operations on an n-set	n^{n^2}	
commutative binary operations	$n^{n(n+1)/2}$	

4. Exam shortcuts and standard counter-examples

Exam shortcut.

- Memorise B_n: 1, 2, 5, 15, 52, 203.
- Equivalence relations on a 4-set: 15; on a 5-set: 52.
- Equivalence relations with exactly two classes on an n-set: $2^{n-1} - 1$.

Exam shortcut.

- The empty relation on a non-empty set is symmetric, antisymmetric, asymmetric and transitive, but *not* reflexive.
- So "symmetric + transitive $\Rightarrow$ reflexive" is false.

Exam shortcut.

- "Symmetric" and "antisymmetric" are not opposites: the identity relation, or $\{(a,a),(b,b)\}$, is both.
- Every asymmetric relation is antisymmetric (and irreflexive).

Exam shortcut.

Standard examples: on $\mathbb{Z}$, $a \sim b \iff n \mid a-b$ is an equivalence relation with n classes forming $\mathbb{Z}_n$; on $\mathbb{R}$, $x \sim y \iff x-y \in \mathbb{Q}$ is an equivalence relation with uncountably many classes; $|x-y| \leq 1$ is reflexive and symmetric but not transitive; $x \leq y$ is a partial order, not an equivalence relation; $a < b$ is not a partial order (not reflexive).

Exam shortcut.

- The number of onto maps is *not* $\binom{m}{n}n!$; use inclusion and exclusion or $n!\,S(m,n)$.
- Transitivity can never be counted by a product rule: use the Bell numbers or the partial-order list.

5. Solved examples

Example 1. Find the number of relations on $\{1,2,3,4\}$ that are both reflexive and antisymmetric.

Solution

- Reflexive forces all 4 diagonal cells to be 1.
- For each of the $\binom{4}{2} = 6$ off-diagonal pairs $\{(a,b),(b,a)\}$, antisymmetry allows three choices: neither, only (a,b), only (b,a).
- Hence the number is $1 \cdot 3^6 = 729$.

Example 2. Find the number of onto functions from a 6-element set to a 3-element set.

Solution

By inclusion and exclusion,

$$3^6 - \binom{3}{1}2^6 + \binom{3}{2}1^6 - 0 = 729 - 192 + 3 = 540.$$

Check: $S(6,3) = 90$ and $3! \cdot 90 = 540$.

Example 3. How many equivalence relations on $\{1,2,3,4,5\}$ contain the pair $(1,2)$? How many have exactly three classes?

Solution

- An equivalence relation containing $(1,2)$ is a partition in which 1 and 2 lie in the same block.
- Glue 1 and 2 into a single element; we then partition a 4-element set, so the answer is $B_4 = 15$.
- Exactly three classes: $S(5,3)$.
- The block sizes are $3+1+1$ ($\binom{5}{3} = 10$ ways) or $2+2+1$ ($\frac{1}{2}\binom{5}{2}\binom{3}{2} = 15$ ways), so $S(5,3) = 25$.

Example 4. How many relations on a 3-element set are both symmetric and antisymmetric? How many are reflexive and symmetric on a 4-element set?

Solution

- Symmetric and antisymmetric together forbid every off-diagonal entry (if $(a,b) \in R$ with $a \neq b$ then $(b,a) \in R$, contradicting antisymmetry).
- The diagonal is free: $2^3 = 8$ relations (the subsets of the identity relation).
- Reflexive and symmetric on 4 points: diagonal fixed, 2 choices per off-diagonal pair, so $2^6 = 64$.

Example 5. Which of the following are equivalence relations? (a) on $\mathbb{R}$, $x \sim y \iff x - y \in \mathbb{Z}$; (b) on $\mathbb{R} \setminus \{0\}$, $x \sim y \iff xy > 0$; (c) on $\mathbb{R}$, $x \sim y \iff xy \geq 0$; (d) on $\mathbb{R}$, $x \sim y \iff x^2 = y^2$.

Solution

- (a) Yes; the classes are the cosets $x + \mathbb{Z}$, one for each point of $[0,1)$.
- (b) Yes: reflexive since $x^2 > 0$; symmetric; transitive since x, y have the same sign and y, z have the same sign.
- Two classes: positives and negatives.
- (c) No: $1 \sim 0$ and $0 \sim -1$, but $1 \cdot (-1) < 0$; transitivity fails.
- (d) Yes; the classes are $\{x, -x\}$. Answer: (a), (b), (d).

Example 6. Let $f : A \to B$, $g : B \to C$ with $g \circ f$ a bijection. Must f or g be a bijection?

Solution

- We only get f one-one and g onto.
- Counter-example: $A = C = \{1\}$, $B = \{1,2\}$, $f(1) = 1$, $g(1) = g(2) = 1$.
- Then $g \circ f$ is the identity of $\{1\}$, but f is not onto and g is not one-one.
- If A, B, C are finite with $|A| = |B| = |C|$, then both are bijections.

Example 7. Let $|A| = 5$, $|B| = 6$, $|A \cap B| = 3$. Find $|P(A) \cap P(B)|$, $|P(A) \cup P(B)|$ and $|(A \times B) \cap (B \times A)|$.

Solution

- $P(A) \cap P(B) = P(A \cap B)$ has $2^3 = 8$ elements.
- Then $|P(A) \cup P(B)| = 32 + 64 - 8 = 88$ (strictly less than $|P(A \cup B)| = 2^8 = 256$).
- Finally $(A \times B) \cap (B \times A) = (A \cap B)^2$ has $3^2 = 9$ elements.